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WO-100-01 — Internal inspection of tank T-101

Area	A100 Tank farm	**Task**	TI Tank internal inspection
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Equipment	T-101	**Lines**	6"-P150-P-1001, 4"-P150-P-6038, 1"-P150-V-6030, 1"-P150-D-6031, 1"-P150-N-3015, 2"-P300-LS-3007, 2"-P150-SC-3008, 3/4"-P150-I-6032, 3/4"-P150-I-6033
Isometrics	ISO-6"-P150-P-1001, ISO-4"-P150-P-6038, ISO-1"-P150-V-6030, ISO-1"-P150-D-6031, ISO-1"-P150-N-3015, ISO-2"-P300-LS-3007, ISO-2"-P150-SC-3008, ISO-3/4"-P150-I-6032, ISO-3/4"-P150-I-6033		
Permits	General work permit, Confined-space entry permit		

Reason

Scheduled internal inspection of the shell, floor and roof.

Scope of work

1. Isolate T-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 15 TMLs; floor scan.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: T-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	6"-P150-P-1001/A2	6"-P150-P-1001	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P150 (6"-P150-P-1001/A2) on T-101-N1	E 32'-2-15/16" N 26'-6" EL 101'-6"		
2	blind	T-101-N1	6"-P150-P-1001	Fit a blind between 6"-P150-P-1001/A2 and the nozzle flange T-101-N1	E 31'-7-13/16" N 26'-6" EL 101'-6"		
3	valve	4"-P150-P-6038/2	4"-P150-P-6038	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-P-6038/2) on T-101-N2	E 25'-0" N 22'-6" EL 123'-10-13/16"		
4	blind	T-101-N2	4"-P150-P-6038	Fit a blind between 4"-P150-P-6038/2 and the nozzle flange T-101-N2	E 25'-0" N 22'-6" EL 123'-3-1/16"		
5	removal	3"-P150-RV-4002/2	3"-P150-RV-4002	Remove the relief valve on T-101-N3 and blind the nozzle			
6	valve	1"-P150-V-6030/2	1"-P150-V-6030	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6030/2) on T-101-N4	E 25'-0" N 26'-6" EL 123'-10-1/2"		
7	blind	T-101-N4	1"-P150-V-6030	Fit a blind between 1"-P150-V-6030/2 and the nozzle flange T-101-N4	E 25'-0" N 26'-6" EL 123'-3-11/16"		
8	valve	1"-P150-D-6031/2	1"-P150-D-6031	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6031/2) on T-101-N5	E 18'-0" N 26'-6" EL 101'-0"		
9	blind	T-101-N5	1"-P150-D-6031	Fit a blind between 1"-P150-D-6031/2 and the nozzle flange T-101-N5	E 18'-6-7/8" N 26'-6" EL 101'-0"		
10	valve	1"-P150-N-3015/B2	1"-P150-N-3015	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3015/B2) on T-101-N6	E 28'-0" N 26'-6" EL 123'-8-1/4"		
11	blind	T-101-N6	1"-P150-N-3015	Fit a blind between 1"-P150-N-3015/B2 and the nozzle flange T-101-N6	E 28'-0" N 26'-6" EL 123'-1-7/16"		
12	valve	2"-P300-LS-3007/B2	2"-P300-LS-3007	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P300 (2"-P300-LS-3007/B2) on T-101-N7	E 18'-5/8" N 26'-6" EL 102'-6"		
13	blind	T-101-N7	2"-P300-LS-3007	Fit a blind between 2"-P300-LS-3007/B2 and the nozzle flange T-101-N7	E 18'-6-1/8" N 26'-6" EL 102'-6"		
14	valve	2"-P150-SC-3008/A2	2"-P150-SC-3008	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3008/A2) on T-101-N8	E 18'-5/8" N 26'-6" EL 104'-6"		
15	blind	T-101-N8	2"-P150-SC-3008	Fit a blind between 2"-P150-SC-3008/A2 and the nozzle flange T-101-N8	E 18'-6-1/8" N 26'-6" EL 104'-6"		
16	valve	3/4"-P150-I-6032/2	3/4"-P150-I-6032	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-I-6032/2) on T-101-N9	E 18'-1/4" N 26'-6" EL 107'-0"		
17	blind	T-101-N9	3/4"-P150-I-6032	Fit a blind between 3/4"-P150-I-6032/2 and the nozzle flange T-101-N9	E 18'-7" N 26'-6" EL 107'-0"		
18	valve	3/4"-P150-I-6033/2	3/4"-P150-I-6033	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-I-6033/2) on T-101-N10	E 18'-1/4" N 26'-6" EL 117'-0"		
19	blind	T-101-N10	3/4"-P150-I-6033	Fit a blind between 3/4"-P150-I-6033/2 and the nozzle flange T-101-N10	E 18'-7" N 26'-6" EL 117'-0"		

Inspections

Id	Inspection	Criteria	Basis
WO-100-01-I1	Internal visual examination of the shell, floor, roof structure and nozzles	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-100-01-I2	Ultrasonic thickness at the 15 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Id	Inspection	Criteria	Basis
WO-100-01-I3	Floor scan for under-floor corrosion over the full floor	Owner's floor acceptance criteria	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-100-01-TML01	T-101	Nozzle neck T-101-N1	E 31'-7-13/16" N 26'-6" EL 101'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML02	T-101	Nozzle neck T-101-N2	E 25'-0" N 22'-6" EL 123'-3-1/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML03	T-101	Nozzle neck T-101-N3	E 22'-0" N 26'-6" EL 123'-4-5/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML04	T-101	Nozzle neck T-101-N4	E 25'-0" N 26'-6" EL 123'-3-11/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML05	T-101	Nozzle neck T-101-N5	E 18'-6-7/8" N 26'-6" EL 101'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML06	T-101	Nozzle neck T-101-N6	E 28'-0" N 26'-6" EL 123'-1-7/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML07	T-101	Nozzle neck T-101-N7	E 18'-6-1/8" N 26'-6" EL 102'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML08	T-101	Nozzle neck T-101-N8	E 18'-6-1/8" N 26'-6" EL 104'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML09	T-101	Nozzle neck T-101-N9	E 18'-7" N 26'-6" EL 107'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML10	T-101	Nozzle neck T-101-N10	E 18'-7" N 26'-6" EL 117'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML11	T-101	Floor centre	E 25'-0" N 26'-6" EL 100'-6"		centre: ____			
WO-100-01-TML12	T-101	Shell at 0.000" above the floor	E 25'-0" N 26'-6" EL 100'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-100-01-TML13	T-101	Shell at 6'-0" above the floor	E 25'-0" N 26'-6" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-100-01-TML14	T-101	Shell at 12'-0" above the floor	E 25'-0" N 26'-6" EL 112'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-100-01-TML15	T-101	Shell at 18'-0" above the floor	E 25'-0" N 26'-6" EL 118'-6"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Service leak test

Description	Every opened joint checked at operating pressure on start-up
Basis	SW-TEST-2

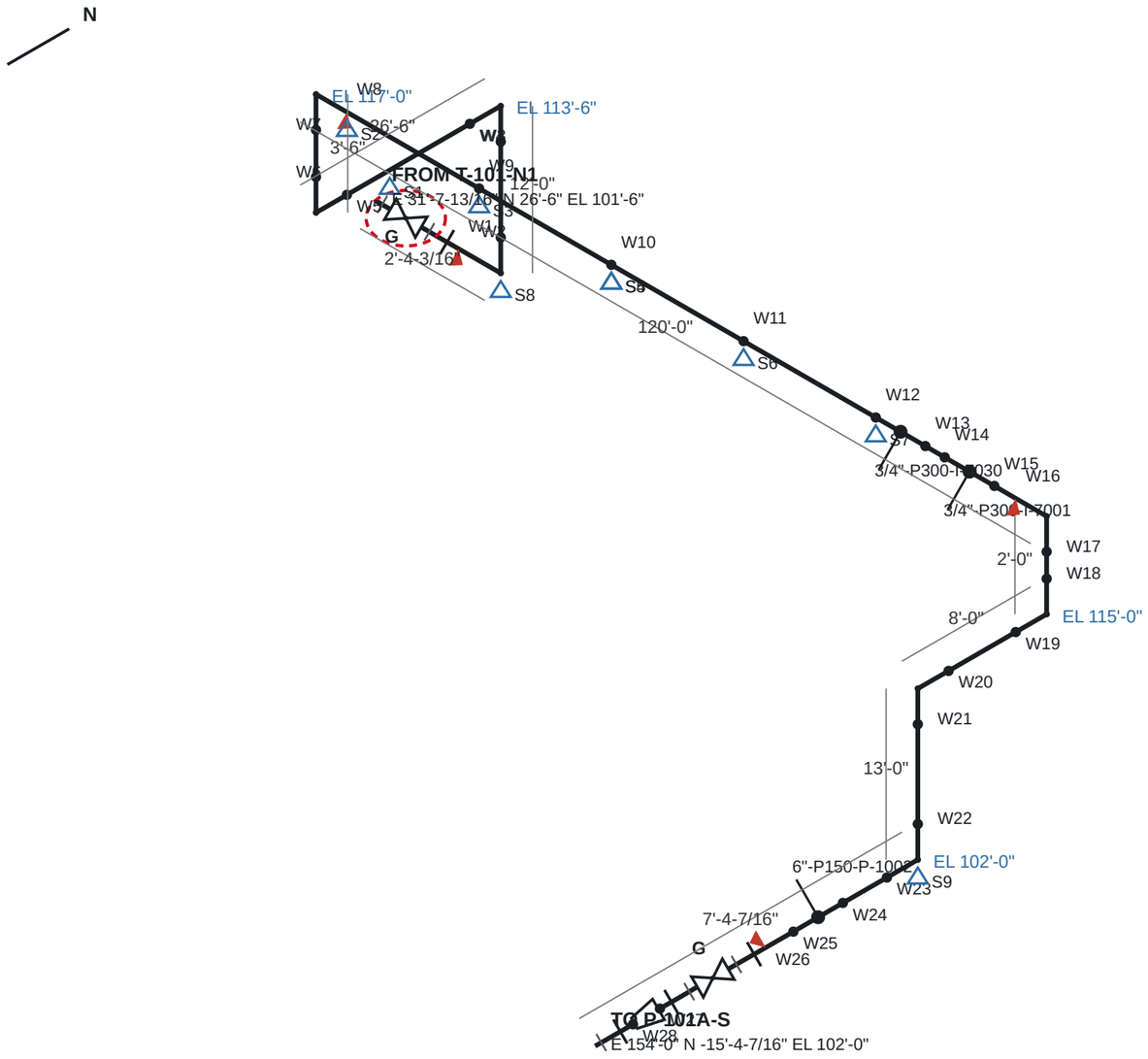
Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Opened for entry; cleaned for inspection	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 15 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Floor scan	SW-WO-6	Owner's floor criteria	Floor scan report	S	R	R				
7	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
8	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
9	Service leak test of opened joints on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
10	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
11	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Isometric ISO-6"-P150-P-1001

ISOMETRIC — 6"-P150-P-1001 — Internal inspection of tank T-101



BILL OF MATERIALS

1	GASKET 6" P150 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P150	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P150	4
4	FLANGE WELD-NECK 6" P150 RAISED FACE	3
5	ELBOW 90° R = 5/3 OD 6" P150	8
6	PIPE 6" P150 BEVELLED ENDS	174'-6-1/8"
7	TEE REDUCING 6" × 3/4" P150	2
8	TEE EQUAL 6" P150	1
9	REDUCER CONCENTRIC 6" × 4" P150	1
10	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1
11	FLANGE WELD-NECK 4" P150 RAISED FACE	1
12	GASKET 4" P150 COMPRESSED SHEET	1

WELDS 28 (4 FIELD) · SUPPORTS 9 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8F W9 W10
W11 W12 W13 W14 W15 W16F W17 W18 W19 W20
W21 W22 W23 W24 W25 W26F W27 W28

6"-P150-P-1001

CLASS H150 — Hydrocarbon process
SERVICE P · 6" · P150 · DESIGN 150 psi
T-101-N1 → P-101A-S
PIPE 174'-6-1/8" · WEIGHT 3040 lb

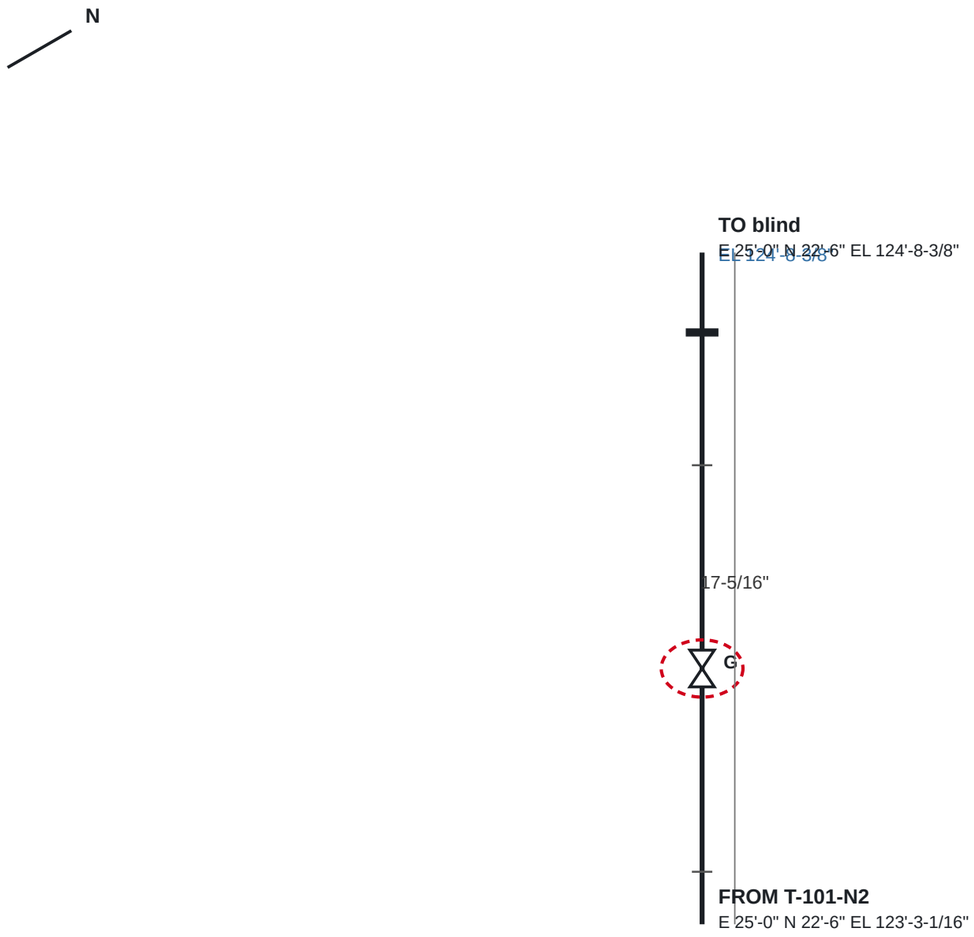
DWG ISO-6"-P150-P-1001 REV 0 WO-100-01

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Isometric ISO-4"-P150-P-6038

ISOMETRIC — 4"-P150-P-6038 — Internal inspection of tank T-101



BILL OF MATERIALS

1	GASKET 4" P150 COMPRESSED SHEET	2
2	VALVE, GATE, WEDGE, OS&Y 4" P150	1
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	2
4	FLANGE BLIND 4" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

4"-P150-P-6038

CLASS H150 — Hydrocarbon process
SERVICE P · 4" · P150 · DESIGN 150 psi
T-101-N2 → blind
PIPE 0.000" · WEIGHT 176 lb

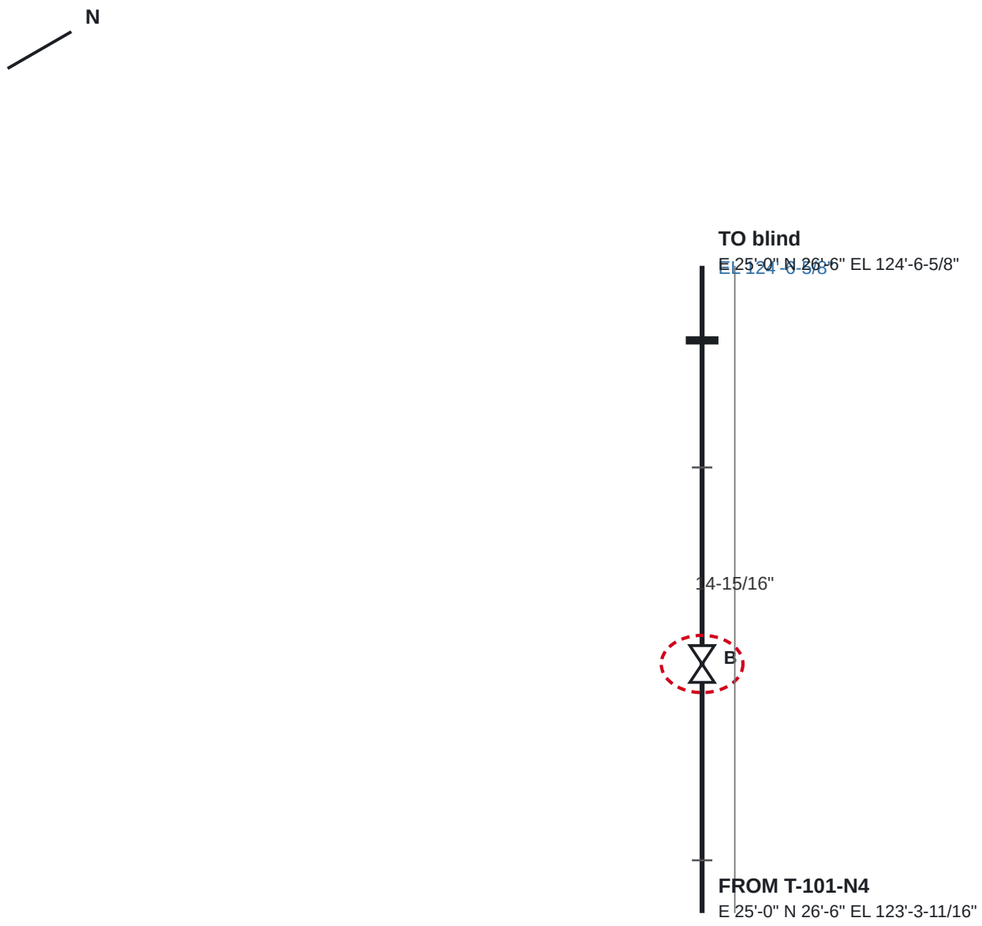
DWG ISO-4"-P150-P-6038 REV 0 WO-100-01

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Isometric ISO-1"-P150-V-6030

ISOMETRIC — 1"-P150-V-6030 — Internal inspection of tank T-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-V-6030

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
T-101-N4 → blind
PIPE 0.000" · WEIGHT 39 lb

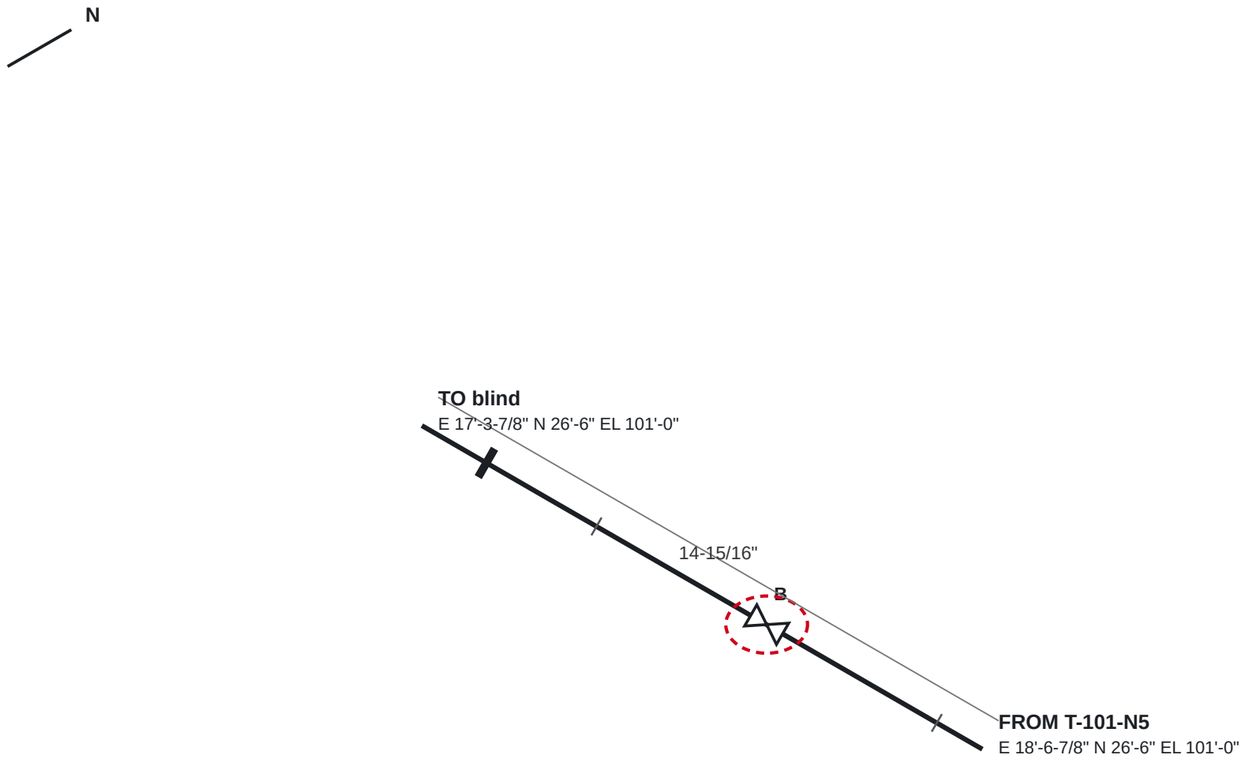
DWG ISO-1"-P150-V-6030 REV 0 WO-100-01

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Isometric ISO-1"-P150-D-6031

ISOMETRIC — 1"-P150-D-6031 — Internal inspection of tank T-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-D-6031

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
T-101-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

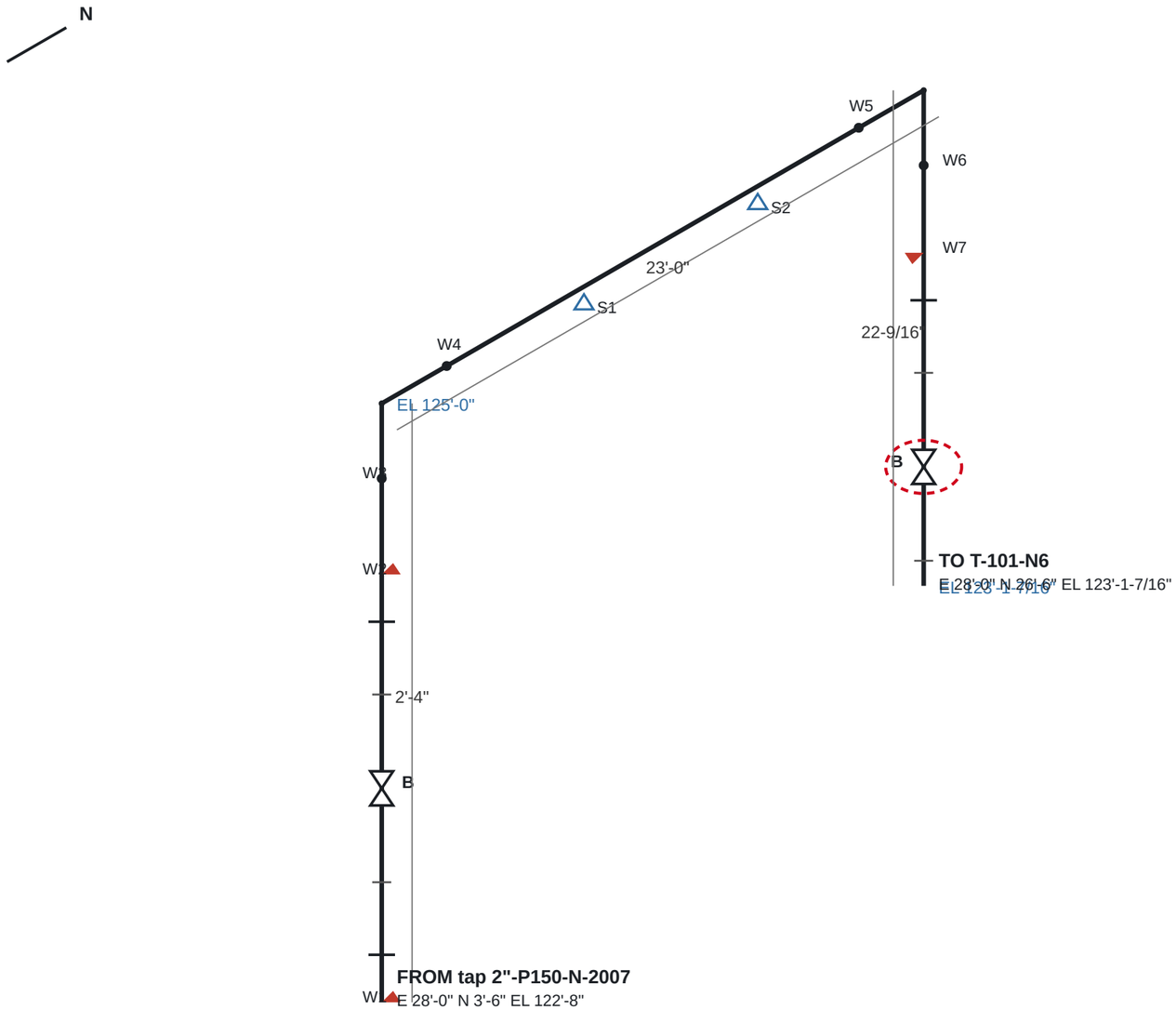
DWG ISO-1"-P150-D-6031 REV 0 WO-100-01

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Isometric ISO-1"-P150-N-3015

ISOMETRIC — 1"-P150-N-3015 — Internal inspection of tank T-101



BILL OF MATERIALS

1	FLANGE WELD-NECK 1" P150 RAISED FACE	3
2	GASKET 1" P150 COMPRESSED SHEET	4
3	VALVE, BALL, FLOATING 1" P150	2
4	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	4
5	PIPE 1" P150 BEVELLED ENDS	23'-4-1/8"
6	ELBOW 90° R = 5/3 OD 1" P150	2

WELDS 7 (3 FIELD) · SUPPORTS 2 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7F

1"-P150-N-3015

CLASS G150 — Air and gases
SERVICE N · 1" · P150 · DESIGN 150 psi
tap 2"-P150-N-2007 → T-101-N6
PIPE 23'-4-1/8" · WEIGHT 104 lb

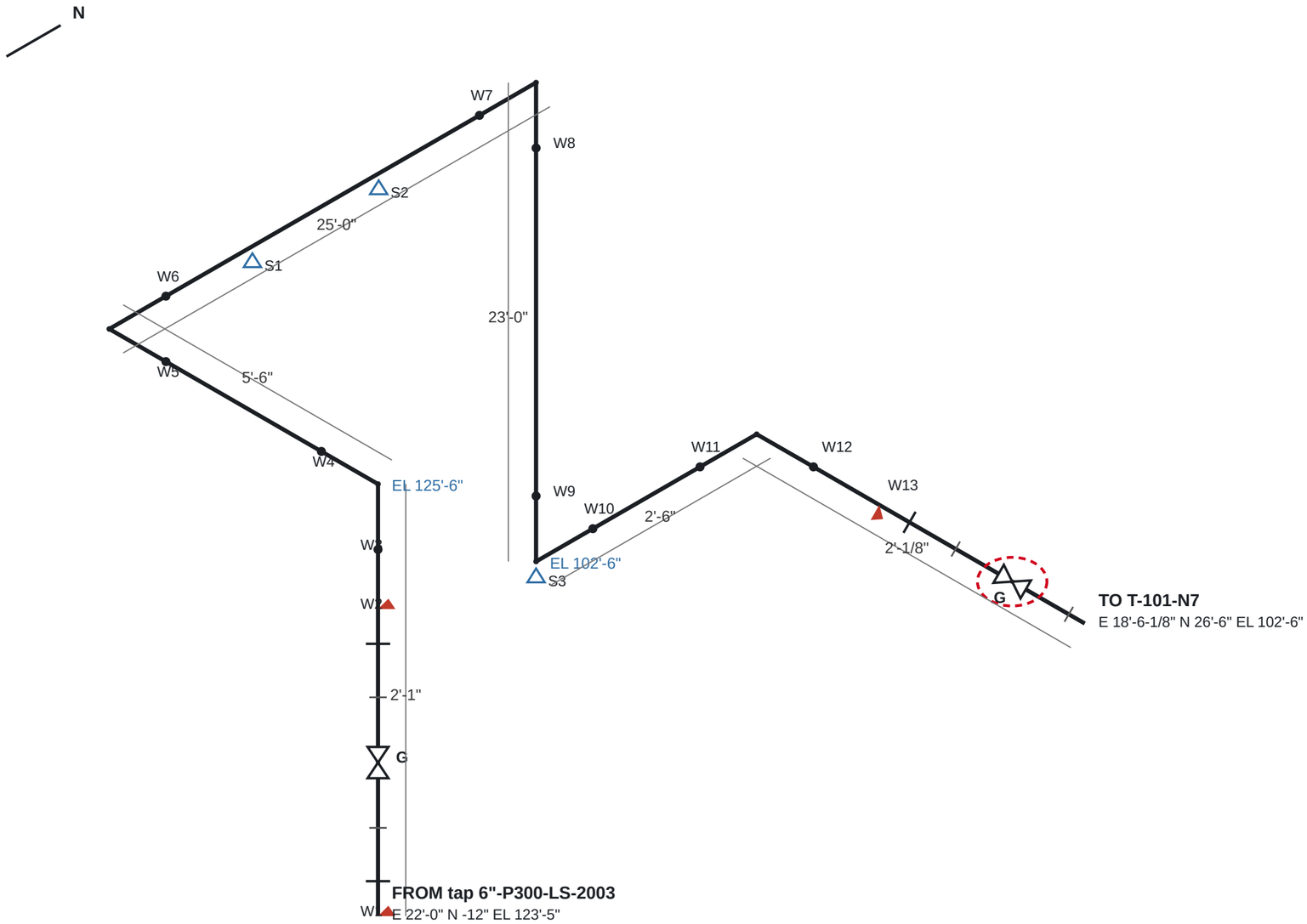
DWG ISO-1"-P150-N-3015 REV 0 WO-100-01

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Isometric ISO-2"-P300-LS-3007

ISOMETRIC — 2"-P300-LS-3007 — Internal inspection of tank T-101



BILL OF MATERIALS

1	FLANGE WELD-NECK 2" P300 RAISED FACE	3
2	GASKET 2" P300 COMPRESSED SHEET	4
3	VALVE, GATE, WEDGE, OS&Y 2" P300	2
4	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300	4
5	PIPE 2" P300 BEVELLED ENDS	54'-5-1/4"
6	ELBOW 90° R = 5/3 OD 2" P300	5

WELDS 13 (3 FIELD) · SUPPORTS 3 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F

2"-P300-LS-3007

CLASS S300 — Low-pressure steam — INSULATED
SERVICE LS · 2" · P300 · DESIGN 300 psi
tap 6"-P300-LS-2003 → T-101-N7
PIPE 54'-5-1/4" · WEIGHT 297 lb

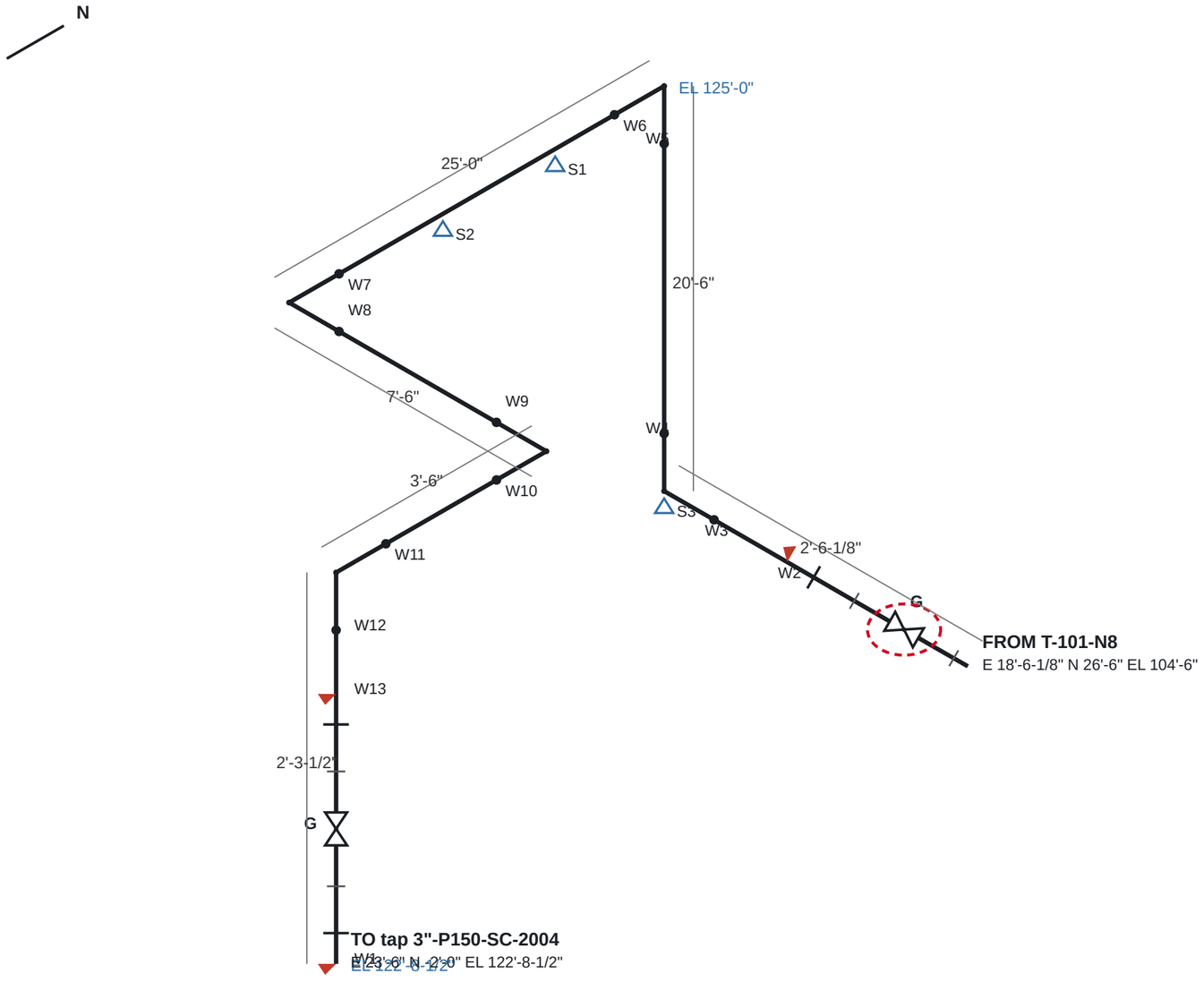
DWG ISO-2"-P300-LS-3007 REV 0 WO-100-01

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Isometric ISO-2"-P150-SC-3008

ISOMETRIC — 2"-P150-SC-3008 — Internal inspection of tank T-101



BILL OF MATERIALS

1	GASKET 2" P150 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 2" P150	2
3	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150	4
4	FLANGE WELD-NECK 2" P150 RAISED FACE	3
5	PIPE 2" P150 BEVELLED ENDS	55'-7-3/4"
6	ELBOW 90° R = 5/3 OD 2" P150	5

WELDS 13 (3 FIELD) · SUPPORTS 3 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F

2"-P150-SC-3008

CLASS C150 — Steam condensate — INSULATED
SERVICE SC · 2" · P150 · DESIGN 150 psi
T-101-N8 → tap 3"-P150-SC-2004
PIPE 55'-7-3/4" · WEIGHT 300 lb

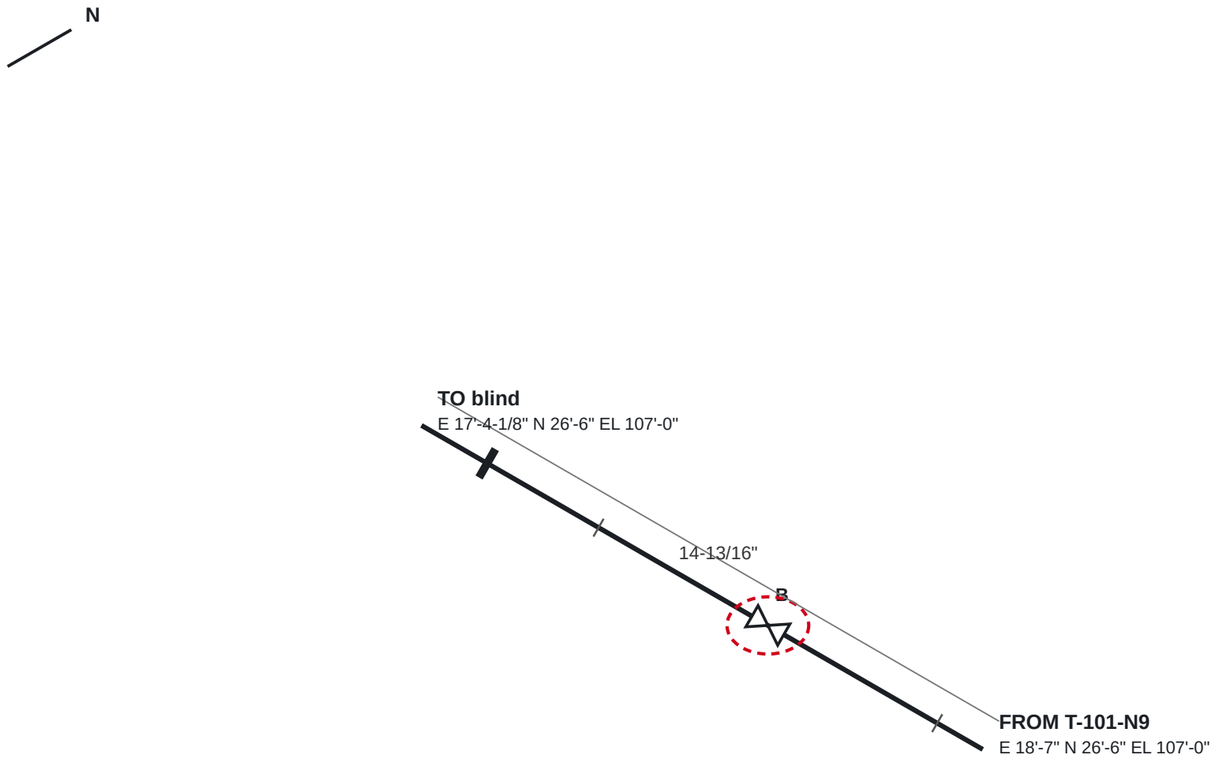
DWG ISO-2"-P150-SC-3008 REV 0 WO-100-01

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Isometric ISO-3/4"-P150-I-6032

ISOMETRIC — 3/4"-P150-I-6032 — Internal inspection of tank T-101



BILL OF MATERIALS

1	GASKET 3/4" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P150	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150	2
4	FLANGE BLIND 3/4" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P150-I-6032

CLASS H150 — Hydrocarbon process
SERVICE I · 0.75" · P150 · DESIGN 150 psi
T-101-N9 → blind
PIPE 0.000" · WEIGHT 32 lb

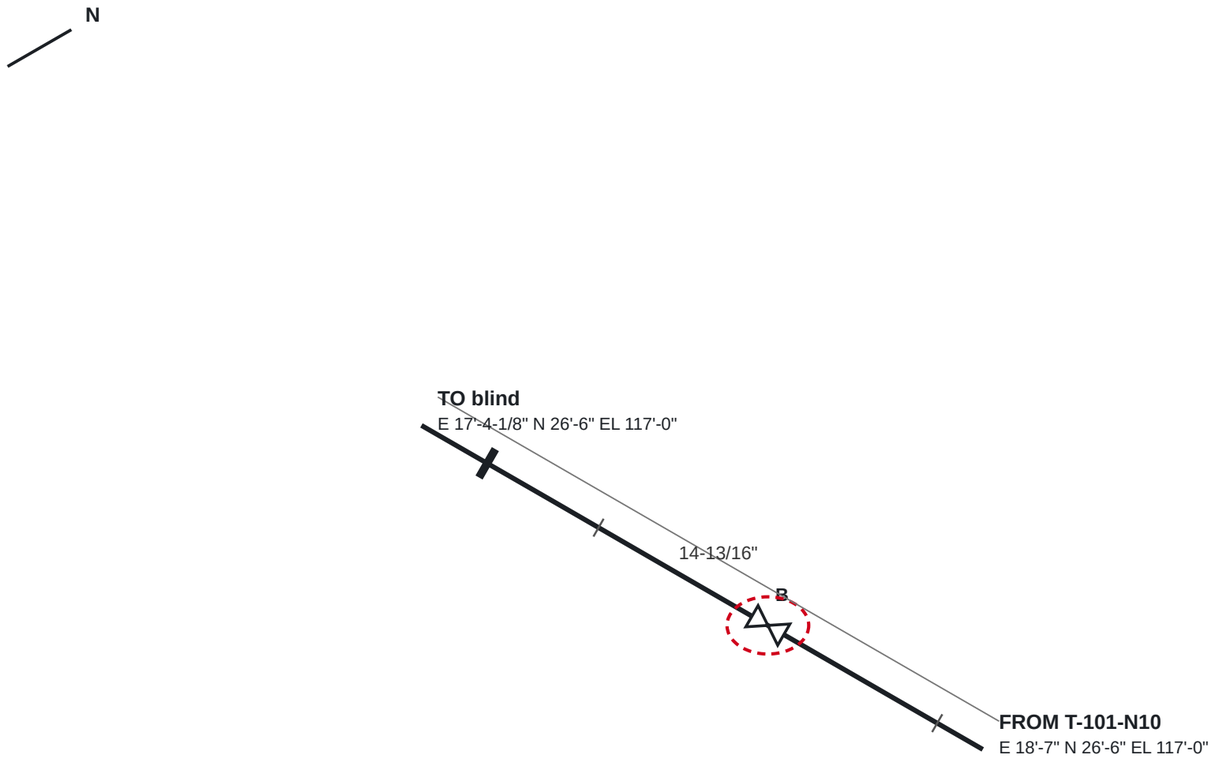
DWG ISO-3/4"-P150-I-6032 REV 0 WO-100-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-3/4"-P150-I-6033

ISOMETRIC — 3/4"-P150-I-6033 — Internal inspection of tank T-101



BILL OF MATERIALS

1	GASKET 3/4" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P150	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150	2
4	FLANGE BLIND 3/4" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P150-I-6033

CLASS H150 — Hydrocarbon process
SERVICE I · 0.75" · P150 · DESIGN 150 psi
T-101-N10 → blind
PIPE 0.000" · WEIGHT 32 lb

DWG ISO-3/4"-P150-I-6033 REV 0 WO-100-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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